Homework 2, 27.10.2026

HOMEWORK 2.5

Exercise 2.5 Let $\Omega = [0,1]$, and let $\mathcal{F}_3$ consist of all countable subsets of Ω, and all subsets of Ω having a countable complement.

a. Show that $\mathcal{F}_3$ is a sigma algebra.

b. Let us define $P(A) = 0$ if A is countable, and $P(A) = 1$ if A has a countable complement. Is $(\Omega, \mathcal{F}_3, P)$ a legitimate probability space?

a) $\mathcal{F}_3 = \{A \subseteq \Omega \mid A \text{ countable OR } A^c \text{ countable}\}$ $\mathcal{F}_3$ is a σ-algebra

i $\emptyset \in \mathcal{F}_3$? ✓ (countable)

ii $A \in \mathcal{F}_3 \Rightarrow A^c \in \mathcal{F}_3$? ✓ $A \in \mathcal{F}_3 \Rightarrow$ A is either countable or A^c is

I. A is countable $\Rightarrow A^c \in \mathcal{F}_3$ $((A^c)^c = A$ is countable

II. A is uncountable $\Rightarrow A^c$ is countable $\Rightarrow A^c \in \mathcal{F}_3$ (def of $\mathcal{F}_3$)

iii $\{A_i\}_{i=1}^{\infty}$ countable, $A_i \in \mathcal{F}_3 \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}_3$?

I. all A_i are countable $\Rightarrow \bigcup_{i=1}^{\infty} A_i$ countable $\in \mathcal{F}_3$ ✓

II. ∃ at least one A_i uncountable $\Rightarrow \bigcup_{i=1}^{\infty} A_i$ uncountable

$\Rightarrow \left(\bigcup_{i=1}^{\infty} A_i\right)^c = \bigcap_{i=1}^{\infty} A_i^c$ ⇝ exists A_j uncountable → since $A_j \in \mathcal{F}_3$ $\Rightarrow A_j^c$ countable $\Rightarrow \left(\bigcap_{i \neq j} A_i^c\right) \cap A_j^c \subseteq A_j^c$ countable ✓

b) $P(A) = 0$ A countable ⇝ Is $(\Omega, \mathcal{F}_3, P)$ probability space?

$P(A) = 1$ A^c countable ⇝ $P(\Omega) = 1$

Both A and A^c cannot be countable: $B = \{A \in \mathcal{F} \mid A \text{ countable}\}$, $C = \{A \in \mathcal{F} \mid A^c \text{ countable}\}$

I. A_i disjoint sets. Let A_i all be countable: $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = 0$ ⇝ since the union is countable, the equation holds

II. let now one (w.l.o.g. A_1) be uncountable: $P\left(\bigcup_{i=1}^{\infty} A_i\right) = 1 + P\left(\bigcup_{i=2} A_i\right) = 1 + 0 = 1$ ⇝ union uncountable, equal to 1 ✓

III. let at least two A_i be uncountable (WLOG A_1, A_2) we want to show $A_1 \cap A_2 \neq \emptyset$ → otherwise the P would be > 1

$A_1 \cap A_2 = (A_1^c \cup A_2^c)^c$ ⇝ now A_1 and A_2 are countable → $A_1 \cup A_2$ is countable

let $B = A_1 \cup A_2$, so $A_1 \cap A_2 = B^c$ ⇒ if $A_1 \cap A_2 = \emptyset$ ⇒ $B = \Omega$, but Ω is uncountable and B is countable

⇒ two uncountable sets cannot be disjoint in $\mathcal{F}_3$ ⇒ $(\Omega, \mathcal{F}_3, P)$ is legitimate p. space ✓